

Replace NA with Mean by Group in R (2 Examples)

In this tutorial you'll learn how to replace NA values with the mean by group in the [R programming language](#).

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Let's take a look.



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Data Sample & Install Packages

Let's say we have a data sample similar to the following one:

```
data_sample <- data.frame(group = c("A","A", "A", "B", "B","B", "C",
                                   x1 = c(1, NA, 3, NA, 2, 4, NA, 7, 1),
                                   x2 = c(NA, 7, 4, -1, NA, 0, 2, NA, 6),
                                   x3 = as.character(c(3, 5, NA, 2, 4, NA, 1,
data_sample
```

	group	x1	x2	x3
1	A	1	NA	3
2	A	NA	7	5
3	A	3	4	NA
4	B	NA	-1	2
5	B	2	NA	4
6	B	4	0	NA
7	C	NA	2	1
8	C	7	NA	0
9	C	1	6	NA

Also, in order to replace the missing values and substitute them by group in our data frame, we will need to install the [dplyr package](#) and the [tidyr package](#), if not installed:

```
install.packages('dplyr')
install.packages('tidyr')
```



Now, load the packages before we start:

```
library(dplyr)
library(tidyr)
```

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Example 1: Replace NA with Mean by Group in all Columns

Now, we can get started. In order to replace our NA values with the mean by group in our data sample, we will use the `group_by()` function from `dplyr` package so that we can divide our data into groups (i.e. A, B, or C).

Next, also from the `dplyr` package, we will use the functions `mutate_at()` and `vars()` so that we can specify the variables we try to modify. Last, we will use the function `replace_na()` from the `tidyr` package, and the mean() function so that we can identify and replace our missing values with the group's mean.

```
data_sample %>%
  group_by(group) %>%
  mutate_at(vars(x1,x2,x3),
            ~replace_na(.,
                        mean(., na.rm = TRUE)))
```

	group	x1	x2	x3
1	A	1	5.5	3.0
2	A	2	7.0	5.0
3	A	3	4.0	4.0
4	B	3	-1.0	2.0
5	B	2	-0.5	4.0
6	B	4	0.0	3.0
7	C	4	2.0	1.0
8	C	7	4.0	0.0
9	C	1	6.0	0.5



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Example 2: Replace NA with Mean by Group in all Numeric Columns

In this example, we will also use the `group_by()` function so that we can divide our data into groups. Next, we can use the `mutate_if()` function from the `dplyr` package in order to replace NA values only in numeric columns. Last, we will use the `ifelse()` function to identify the missing values and replace them with the mean by group.

```
data_sample %>%
  group_by(group) %>%
  mutate_if(is.numeric,
            function(x) ifelse(is.na(x),
                               mean(x, na.rm = TRUE),
                               x))
```

	group	x1	x2	x3
1	A	1	5.5	3.0
2	A	2	7.0	5.0
3	A	3	4.0	4.0
4	B	3	-1.0	2.0
5	B	2	-0.5	4.0
6	B	4	0.0	3.0
7	C	4	2.0	1.0
8	C	7	4.0	0.0
9	C	1	6.0	0.5

As shown, in order to handle missing values in our data frame, we can replace them with the mean by group if our data is divided by groups, in all columns or exclusively in numeric columns.



Video, Further Resources & Summary

Do you need more explanations on how to replace the missing values with averages by group in R? Then you should have a look at the following YouTube video of the Statistics Globe YouTube channel.

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The YouTube video will be added soon.

Furthermore, you could have a look at some of the related tutorials on Statistics Globe:

- [Remove NA Values in Only One Column of Data Frame in R](#)
- [Replace NA Values by Row Mean in R](#)
- [R Error missing values are not allowed](#)
- [Replace NA by FALSE in R](#)

This post has shown how to **replace NA values with the average by group in R**. In case you have further questions, you may leave a comment below.



This page was created in collaboration with Paula Villasante Soriano. Please have a look at [Paula's author page](#) to get more information about her academic background and the other articles she has written for Statistics Globe.

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